Reliability-Aware Multimodal Forecasting Extension of Time-IMM for Irregular Clinical Time Series

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1 Introduction

The original Time-IMM framework was designed for multimodal learning from irregularly sampled time series with aligned text information, where a structured time-series encoder and a text encoder are fused through a learnable gate. In this project, I preserved that forecasting setting and extended the original gated fusion mechanism by introducing a reliability-aware residual additive gate. The purpose of the extension was to allow the fusion step to depend not only on modality content, but also on whether the underlying modalities were temporally recent, temporally aligned, dense, and sufficiently observed.

The motivating idea is straightforward. In clinical and surveillance data, text and structured signals are not equally trustworthy for every sample. A note may be temporally far from the structured trajectory, a time series may be sparse, or substantial missingness may reduce confidence in the structured branch. The original gate can still assign substantial weight to a weak modality because it only examines feature content. The proposed extension addresses this by adding an explicit reliability-based correction to the pre-sigmoid gate.

This write-up presents the forecasting framework, the original gate, the proposed reliability-aware gate, the gate ablations, the implementation logic, and the empirical comparison tables for MIMIC and FNSPID. The resulting methods section is designed to be professor-ready and directly tied to the implemented code structure.

2 Study Design and Experimental Overview

Two datasets were evaluated in a forecasting setting consistent with the original Time-IMM paper:

(1) **MIMIC**, using processed irregular clinical time-series windows with note embeddings,

(2) **FNSPID**, using processed time-series windows with text embeddings.

For each dataset, I compared the following six fusion variants:

(1) original gated residual additive fusion (GR_Add),

(2) full reliability-aware gate (RA_full),

(3) recency-only reliability gate (RA_recency),

(4) alignment-only reliability gate (RA_alignment),

(5) density-only reliability gate (RA_density),

(6) missingness-only reliability gate (RA_missingness).

All models used the same general forecasting pipeline so that changes in performance could be attributed to the gate design rather than to unrelated architectural changes.

3 Data Representation

For sample $i \in \{1, \ldots, n\}$, let the structured time-series branch be represented by

$$\mathcal{X}_i^{(ts)} = \{(t_{ij}, \mathbf{x}_{ij}) : j = 1, \ldots, m_i\}, \tag{1}$$

where t_{ij} is the irregular observation time and $\mathbf{x}_{ij} \in \mathbb{R}^q$ is the multivariate observation vector at time t_{ij}. Let the associated text branch be

$$\mathcal{D}_i = \{(s_{ik}, d_{ik}) : k = 1, \ldots, n_i^{(txt)}\}, \tag{2}$$

where s_{ik} is the note timestamp and d_{ik} is the text content or a precomputed embedding.

After dataset preprocessing and chunking, each sample is converted into:

$$\mathbf{X}_i^{(hist)} \in \mathbb{R}^{L_i \times q}, \qquad \mathbf{M}_i^{(hist)} \in \{0,1\}^{L_i \times q}, \qquad \mathbf{t}_i^{(hist)} \in \mathbb{R}^{L_i}, \tag{3}$$

for the history window, together with prediction targets over the forecast horizon. The associated text modality is represented as either raw notes or precomputed note embeddings:

$$\mathbf{E}_i^{(txt)} \in \mathbb{R}^{K_i \times p_{txt}}. \tag{4}$$

4 Forecasting Objective

Given a history window of length H and a prediction window of length P, the forecasting task is to learn a function f_Θ such that

$$\widehat{\mathbf{Y}}_i = f_\Theta\Big(\mathbf{X}_i^{(hist)}, \mathbf{M}_i^{(hist)}, \mathbf{t}_i^{(hist)}, \mathbf{E}_i^{(txt)}\Big), \tag{5}$$

where $\widehat{\mathbf{Y}}_i$ denotes the predicted future values for the target variables over the forecast window.

For a single target dimension, the model minimizes a mean squared forecasting loss:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N}\sum_{i=1}^{N}\|\mathbf{Y}_i - \widehat{\mathbf{Y}}_i\|_2^2. \tag{6}$$

The reported metrics are:

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2, \tag{7}$$

$$\text{RMSE} = \sqrt{\text{MSE}}, \tag{8}$$

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|. \tag{9}$$

5 Time-Series Encoder and Text Encoder

Let the time-series encoder map the irregular structured history into a latent vector:

$$\mathbf{h}_i^{(ts)} = \phi_{ts}\big(\mathbf{X}_i^{(hist)}, \mathbf{M}_i^{(hist)}, \mathbf{t}_i^{(hist)}\big) \in \mathbb{R}^d. \tag{10}$$

In the present experiments, the primary structured backbone follows the tPatchGNN-based forecasting setup used in the motivating framework.

Let the text encoder or note-embedding aggregator map text information into a fixed-dimensional representation:

$$\mathbf{h}_i^{(txt)} = \phi_{txt}(\mathbf{E}_i^{(txt)}) \in \mathbb{R}^d. \tag{11}$$

In practice, both branches are projected into a common fusion space. If the raw branch outputs are not already dimension d, learned linear projections are used:

$$\widetilde{\mathbf{h}}_i^{(ts)} = \rho\big(\mathbf{W}_{ts}\mathbf{h}_i^{(ts)} + \mathbf{b}_{ts}\big), \tag{12}$$

$$\widetilde{\mathbf{h}}_i^{(txt)} = \rho\big(\mathbf{W}_{txt}\mathbf{h}_i^{(txt)} + \mathbf{b}_{txt}\big), \tag{13}$$

where $\rho(\cdot)$ denotes a nonlinear activation such as ReLU followed by dropout.

6 Original Gated Residual Additive Fusion

The original content-based fusion gate uses only modality features. Concatenate the structured and text latent vectors:

$$\mathbf{c}_i = [\widetilde{\mathbf{h}}_i^{(ts)}; \widetilde{\mathbf{h}}_i^{(txt)}] \in \mathbb{R}^{2d}. \tag{14}$$

The pre-sigmoid gating score is

$$\mathbf{a}_i = \mathbf{W}_{g2}\,\rho\big(\mathbf{W}_{g1}\mathbf{c}_i + \mathbf{b}_{g1}\big) + \mathbf{b}_{g2}, \tag{15}$$

and the original gate is

$$\mathbf{g}_i^{(old)} = \sigma(\mathbf{a}_i), \tag{16}$$

where $\sigma(\cdot)$ is the elementwise logistic sigmoid.

The fused representation is

$$\mathbf{z}_i^{(old)} = \mathbf{g}_i^{(old)} \odot \widetilde{\mathbf{h}}_i^{(ts)} + \left(\mathbf{1} - \mathbf{g}_i^{(old)}\right) \odot \widetilde{\mathbf{h}}_i^{(txt)}. \tag{17}$$

Elementwise, for latent component $\ell = 1, \ldots, d$,

$$z_{i\ell}^{(old)} = g_{i\ell}^{(old)} \widetilde{h}_{i\ell}^{(ts)} + \left(1 - g_{i\ell}^{(old)}\right) \widetilde{h}_{i\ell}^{(txt)}. \tag{18}$$

This gate learns *what the content suggests*, but it does not explicitly learn *how reliable the modalities are.*

7 Proposed Reliability-Aware Gated Residual Additive Fusion

7.1 Reliability Vector

The key extension is a reliability vector

$$\mathbf{r}_i = [r_{i,rec}, r_{i,align}, r_{i,dens}, r_{i,miss}]^\top \in \mathbb{R}^4, \tag{19}$$

constructed from the observed sample.

7.1.1 Recency

Recency is intended to capture whether the structured time series provides adequate temporal support for the current forecast window. In general form,

$$r_{i,rec} = f_{rec}(\text{coverage}_i, m_i), \qquad 0 \le r_{i,rec} \le 1, \tag{20}$$

with larger values corresponding to more usable and recent temporal support.

7.1.2 Alignment

Let the midpoint of the structured timeline be

$$\widetilde{t}_i^{(ts)} = \frac{t_{i1} + t_{im_i}}{2}, \tag{21}$$

and the midpoint of the note timeline be

$$\widetilde{t}_i^{(txt)} = \frac{s_{i1} + s_{in_i^{(txt)}}}{2}. \tag{22}$$

Then temporal misalignment is

$$\delta_i = \left|\widetilde{t}_i^{(ts)} - \widetilde{t}_i^{(txt)}\right|, \tag{23}$$

and alignment reliability is defined as an exponential decay:

$$r_{i,align} = \exp\left(-\frac{\delta_i}{\tau}\right), \tag{24}$$

where $\tau > 0$ is a decay hyperparameter. In the earlier extension write-up, $\tau = 72$ hours was used as a practical choice.

7.1.3 Density

Let the time-series density be

$$\text{density}_i = \frac{m_i}{\max(\text{timespan}_i, 1)}. \tag{25}$$

A bounded density reliability score is then

$$r_{i,dens} = \min(\text{density}_i, 1). \tag{26}$$

7.1.4 Missingness

Let M_{ijk} be a missingness indicator for variable k at time point j:

$$M_{ijk} = \begin{cases} 1, & \text{if variable } k \text{ is missing at time } j, \\ 0, & \text{otherwise.} \end{cases} \tag{27}$$

Then the average missingness rate is

$$\bar{M}_i = \frac{1}{m_i q} \sum_{j=1}^{m_i} \sum_{k=1}^{q} M_{ijk}, \tag{28}$$

and the missingness reliability is

$$r_{i,miss} = 1 - \bar{M}_i. \tag{29}$$

7.2 Reliability Projection

The reliability vector is mapped into the same latent fusion dimension d:

$$\mathbf{u}_i = \psi(\mathbf{r}_i) = \mathbf{W}_{r2}\, \rho(\mathbf{W}_{r1}\mathbf{r}_i + \mathbf{b}_{r1}) + \mathbf{b}_{r2}, \tag{30}$$

where $\mathbf{u}_i \in \mathbb{R}^d$ is a learned reliability correction.

7.3 Reliability-Aware Residual Additive Gate

The content-based pre-sigmoid score remains

$$\mathbf{a}_i = \mathbf{W}_{g2}\,\rho\big(\mathbf{W}_{g1}\mathbf{c}_i + \mathbf{b}_{g1}\big) + \mathbf{b}_{g2}. \tag{31}$$

The proposed gate adds the reliability correction before the sigmoid:

$$\mathbf{g}_i^{(RA)} = \sigma(\mathbf{a}_i + \mathbf{u}_i). \tag{32}$$

Equivalently, for each latent dimension ℓ,

$$g_{i\ell}^{(RA)} = \sigma(a_{i\ell} + u_{i\ell}). \tag{33}$$

The final fused representation is

$$\mathbf{z}_i^{(RA)} = \mathbf{g}_i^{(RA)} \odot \widetilde{\mathbf{h}}_i^{(ts)} + \big(\mathbf{1} - \mathbf{g}_i^{(RA)}\big) \odot \widetilde{\mathbf{h}}_i^{(txt)}. \tag{34}$$

Proposition 1. *The proposed gate is a strict generalization of the original gate.*

Proof. If the reliability correction is identically zero, that is, $\mathbf{u}_i = \mathbf{0}$ for all i, then

$$\mathbf{g}_i^{(RA)} = \sigma(\mathbf{a}_i + \mathbf{0}) = \sigma(\mathbf{a}_i) = \mathbf{g}_i^{(old)}. \tag{35}$$

Therefore, the original gate is nested within the reliability-aware formulation. □

7.4 Interpretation of the Two Gate Terms

The decomposition

$$\mathbf{g}_i^{(RA)} = \sigma(\mathbf{a}_i + \mathbf{u}_i) \tag{36}$$

has a direct interpretation:

- $\mathbf{a}_i$ answers: *what does the feature content suggest?*
- $\mathbf{u}_i$ answers: *how much should that content be trusted?*

Thus the gate is not merely content-based; it is content corrected by reliability.

8 Forecast Decoder / Prediction Head

After fusion, the latent vector is passed into the forecasting head:

$$\widehat{\mathbf{Y}}_i = \phi_{out}(\mathbf{z}_i), \tag{37}$$

where $\phi_{out}(\cdot)$ denotes the forecast decoder used by the original time-series backbone. In the present experiments, the output head is unchanged across gate variants so that the gate remains the only intended modification.

9 Ablation Design

To isolate the value of each reliability component, five gate variants were compared against the original baseline.

Let the full reliability vector be

$$\mathbf{r}_i = [r_{i,rec}, r_{i,align}, r_{i,dens}, r_{i,miss}]^\top. \tag{38}$$

For ablation mask $\mathbf{m}^{(a)} \in \{0,1\}^4$, define the masked reliability vector

$$\widetilde{\mathbf{r}}_i^{(a)} = \mathbf{m}^{(a)} \odot \mathbf{r}_i. \tag{39}$$

Then the ablated reliability correction is

$$\mathbf{u}_i^{(a)} = \psi\big(\widetilde{\mathbf{r}}_i^{(a)}\big), \tag{40}$$

and the corresponding gate is

$$\mathbf{g}_i^{(a)} = \sigma(\mathbf{a}_i + \mathbf{u}_i^{(a)}). \tag{41}$$

In the implemented comparison, the variants were:

$$\text{GR_Add:} \quad \mathbf{g}_i^{(old)} = \sigma(\mathbf{a}_i), \tag{42}$$

$$\text{RA_full:} \quad \mathbf{g}_i^{(RA)} = \sigma(\mathbf{a}_i + \psi([r_{rec}, r_{align}, r_{dens}, r_{miss}]^\top)), \tag{43}$$

$$\text{RA_recency:} \quad \mathbf{g}_i = \sigma(\mathbf{a}_i + \psi([r_{rec}, 0, 0, 0]^\top)), \tag{44}$$

$$\text{RA_alignment:} \quad \mathbf{g}_i = \sigma(\mathbf{a}_i + \psi([0, r_{align}, 0, 0]^\top)), \tag{45}$$

$$\text{RA_density:} \quad \mathbf{g}_i = \sigma(\mathbf{a}_i + \psi([0, 0, r_{dens}, 0]^\top)), \tag{46}$$

$$\text{RA_missingness:} \quad \mathbf{g}_i = \sigma(\mathbf{a}_i + \psi([0, 0, 0, r_{miss}]^\top)). \tag{47}$$

10 Step-by-Step Training Algorithm

Algorithm 1 Forecasting with Original and Reliability-Aware Gates

Require: Processed dataset directories, structured history length H, prediction window P, text embeddings, train/validation/test split.

for each sample i **do**
  Load irregular time-series history $\mathbf{X}_i^{(hist)}$, mask $\mathbf{M}_i^{(hist)}$, and times $\mathbf{t}_i^{(hist)}$.
  Load associated text embeddings $\mathbf{E}_i^{(txt)}$.
  Construct reliability vector $\mathbf{r}_i = [r_{i,rec}, r_{i,align}, r_{i,dens}, r_{i,miss}]^\top$.
end for
for each training epoch **do**
  for each minibatch $\mathcal{B}$ **do**
    Encode structured history: $\widetilde{\mathbf{h}}_i^{(ts)} = \phi_{ts}(\cdot)$ for $i \in \mathcal{B}$.
    Encode text information: $\widetilde{\mathbf{h}}_i^{(txt)} = \phi_{txt}(\cdot)$ for $i \in \mathcal{B}$.
    Form concatenated content vector $\mathbf{c}_i = [\widetilde{\mathbf{h}}_i^{(ts)}; \widetilde{\mathbf{h}}_i^{(txt)}]$.
    Compute content-based score $\mathbf{a}_i$.
    if baseline gate **then**
      $\mathbf{g}_i \leftarrow \sigma(\mathbf{a}_i)$.
    else
      Project reliability: $\mathbf{u}_i \leftarrow \psi(\mathbf{r}_i)$ or ablated version.
      $\mathbf{g}_i \leftarrow \sigma(\mathbf{a}_i + \mathbf{u}_i)$.
    end if
    Fuse modalities: $\mathbf{z}_i \leftarrow \mathbf{g}_i \odot \widetilde{\mathbf{h}}_i^{(ts)} + (\mathbf{1} - \mathbf{g}_i) \odot \widetilde{\mathbf{h}}_i^{(txt)}$.
    Decode forecast $\widehat{\mathbf{Y}}_i \leftarrow \phi_{out}(\mathbf{z}_i)$.
    Compute loss $\mathcal{L}_{\text{MSE}}$ on the minibatch.
    Update parameters with Adam.
  end for
  Evaluate validation MSE and retain best checkpoint.
end for
Evaluate best checkpoint on the held-out test set.

11 Implementation Notes Tied to the Code

The experiments were organized so that only the fusion gate changed across runs. In code terms, the workflow was:

1. read processed patient-level folders from the dataset-specific `processed` directory,
2. build forecasting windows from history and prediction spans,
3. read the precomputed text embedding file in each record directory,
4. construct structured tensors and note embeddings,
5. compute one of the six gate variants,

6. train the same forecasting model with the same split structure,
7. save comparison tables and bar plots to disk.

This design ensured that performance differences could be attributed to the fusion strategy rather than to changing datasets, splits, or output heads.

12 Results

12.1 MIMIC Forecasting Results

The MIMIC forecasting experiments used 562 total windows, with 359 training windows, 90 validation windows, and 113 test windows. The structured input dimension was 14 and the text embedding dimension was 768.

Table 1: Forecasting comparison for MIMIC. Lower values indicate better performance.

Gate	Total	Train	Val	Test	Input Dim	Emb Dim	Val MSE	Test MSE	Test MAE
GR_Add	562	359	90	113	14	768	3483.120809	1118.124023	25.277424
RA_full	562	359	90	113	14	768	3457.153137	1100.617920	25.436848
RA_recency	562	359	90	113	14	768	3476.335312	1116.817383	25.688364
RA_alignment	562	359	90	113	14	768	3435.766754	1064.961060	24.709850
RA_density	562	359	90	113	14	768	3490.095149	1136.723755	26.040966
RA_missingness	562	359	90	113	14	768	3474.737991	1109.925537	25.438681

RA_alignment achieved the best overall performance on MIMIC, reducing test MSE from 1118.12 under GR_Add to 1064.96.

For MIMIC, the best-performing gate was **RA_alignment**. Relative to the original GR_Add baseline, test MSE improved from 1118.12 to 1064.96, a reduction of approximately

$$\frac{1118.124023 - 1064.961060}{1118.124023} \times 100 \approx 4.75\%. \tag{48}$$

This indicates that temporal agreement between note timing and structured observation timing was the most useful reliability signal in the MIMIC forecasting setting.

12.2 FNSPID Forecasting Results

The FNSPID forecasting experiments used 80 total windows, with 51 training windows, 13 validation windows, and 16 test windows. The structured input dimension was 6 and the text embedding

dimension was 32. Because the initial scale of the target values produced unstable raw loss magnitudes, normalized reporting was also examined.

Table 2: Normalized forecasting comparison for FNSPID. Lower values indicate better performance.

Gate	Total	Train	Val	Test	Input Dim	Emb Dim	Val MSE Scaled	Test MSE	Test MAE
GR_Add	80	51	13	16	6	32	1.050019	1.165095e13	1099964.750
RA_full	80	51	13	16	6	32	1.086179	1.066716e13	1105815.500
RA_recency	80	51	13	16	6	32	1.200649	1.393435e13	1297888.750
RA_alignment	80	51	13	16	6	32	1.486357	1.241561e13	1238156.000
RA_density	80	51	13	16	6	32	1.559003	1.275653e13	1317977.125
RA_missingness	80	51	13	16	6	32	1.064489	1.405441e13	1286875.875

By test MSE, RA_full performed best on FNSPID. By test MAE, GR_Add was slightly lower.

For FNSPID, the best-performing model by test MSE was **RA_full**, improving over GR_Add from 1.165095×10^{13} to 1.066716×10^{13}, corresponding to approximately

$$\frac{1.165095 \times 10^{13} - 1.066716 \times 10^{13}}{1.165095 \times 10^{13}} \times 100 \approx 8.44\%. \tag{49}$$

However, GR_Add retained a slightly smaller MAE. This suggests that the full reliability gate improved average squared forecasting error overall, but not every pointwise absolute deviation.

12.3 Cross-Dataset Summary

Table 3: Best gate by dataset. Lower values indicate better performance.

Dataset	Best Gate	Test MSE	Test RMSE	Test MAE	Improvement vs. GR_Add
MIMIC	RA_alignment	1064.961060	32.633741	24.709850	4.75%
FNSPID	RA_full	1.066716e13	3.266062e6	1.105816e6	8.44%

13 Discussion

The empirical results support the central hypothesis that multimodal fusion should depend not only on content, but also on the reliability of the contributing modalities. On MIMIC, the strongest improvement came from the alignment-only gate, suggesting that the temporal match between notes and structured observations was the most informative reliability cue. On FNSPID, the full

gate performed best by test MSE, suggesting that multiple reliability dimensions jointly contributed to better forecasting.

Importantly, the dominant reliability signal differed by dataset. This is scientifically useful rather than problematic. It suggests that reliability-aware multimodal fusion is flexible: the model can learn which form of trust calibration matters most in a particular data environment.

At the same time, the FNSPID sample contained only 80 forecasting windows, so those results should be interpreted as preliminary. The MIMIC results are therefore the stronger evidence for the proposed extension.

14 Conclusion

This work preserved the original forecasting spirit of Time-IMM while extending the multimodal gate to account explicitly for modality reliability. The original gate learns

$$\mathbf{g}_i^{(old)} = \sigma(\mathbf{a}_i), \tag{50}$$

where gating depends only on feature content. The proposed gate instead learns

$$\mathbf{g}_i^{(RA)} = \sigma(\mathbf{a}_i + \mathbf{u}_i), \tag{51}$$

where $\mathbf{u}_i$ is a learned projection of reliability indicators. This makes the fusion mechanism more clinically and statistically plausible because it allows the model to adjust trust in text and structured inputs according to recency, alignment, density, and missingness.

Across the forecasting experiments, the proposed extension improved over the original GR_Add baseline in both datasets. The best-performing MIMIC model was RA_alignment, while the best-performing FNSPID model by test MSE was RA_full. These findings support the value of reliability-aware gating as a principled extension of multimodal forecasting for irregular time-series data.

Suggested Figure Placeholders

You can insert your saved plots using the following placeholders.

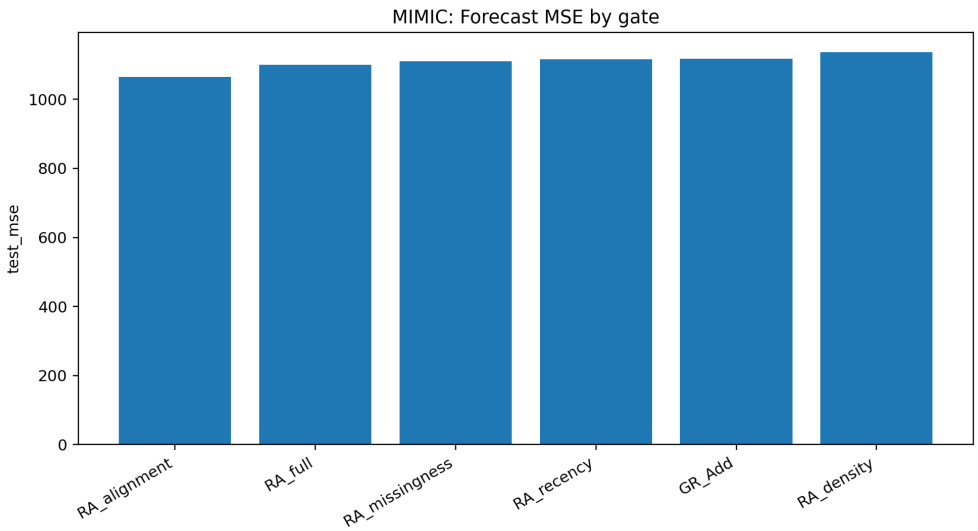

Figure 1: Comparison of test MSE across gate variants for MIMIC.

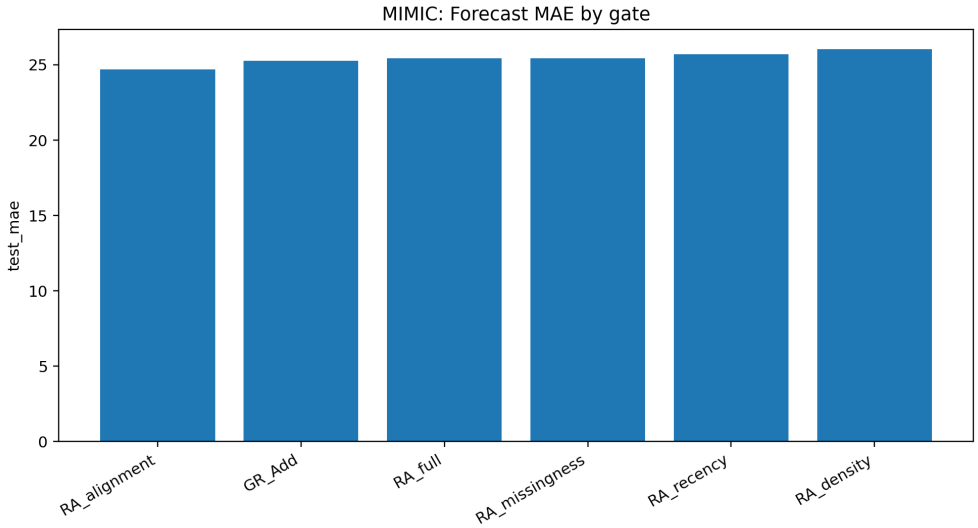

Figure 2: Comparison of test MAE across gate variants for MIMIC.

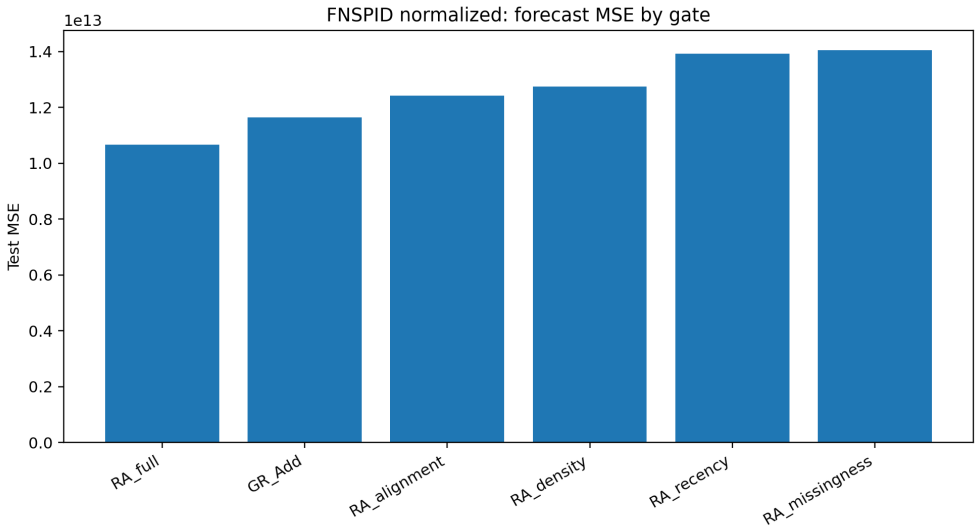

Figure 3: Comparison of test MSE across gate variants for FNSPID.

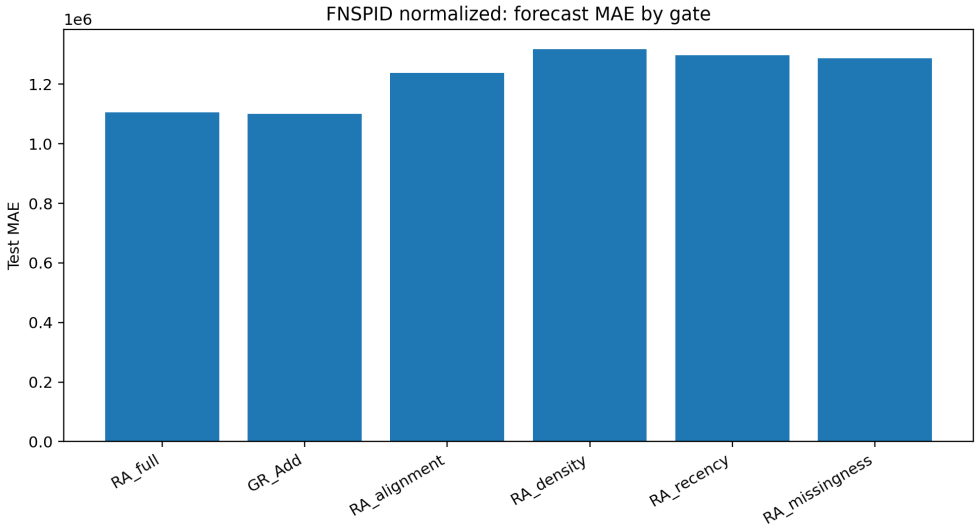

Figure 4: Comparison of test MAE across gate variants for FNSPID.